The Derivative as a Limit and an Instantaneous Rate

Directions:

- The definition in use throughout: $f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}$. The quotient before the limit is an *average* rate over a window of width h; the limit is the *instantaneous* rate at a.
- Where a table appears, fill it in first. The table is the numerical picture of the limit — it shows the average rates closing in on a value; the algebra then proves what that value is.
- Simplify the difference quotient *before* letting $h \to 0$; substituting $h = 0$ first gives $0/0$, which says nothing.

1. Let $f(x) = x^2$. Complete the table of difference quotients at $a = 4$, then find $f'(4)$ from the definition by simplifying the quotient and taking the limit.

h	1	0.5	0.1	0.01
$\frac{f(4+h) - f(4)}{h}$				

Answer: ________________

2. A cart's position is $s(t) = t^2 + 2t$ metres at time t seconds. Complete the average-velocity table on the windows starting at $t = 2$, then compute the instantaneous velocity at $t = 2$ from the limit definition.

window width h	1	0.5	0.1	0.01
average velocity				

Answer: ________________ meters per second

3. A tank holds $V(t) = t^3 - 6t^2 + 9t$ gallons of water t minutes after a valve opens. Find $V'(t)$, and write one sentence saying what $V'(t)$ measures about the tank — not what it equals, but what it tells you.

Answer: ________________

4. Using your $V'(t)$ from problem 3, find the instantaneous rate at which the tank's volume is changing at $t = 5$ minutes. State whether the tank is filling or draining at that instant, and how you know from the sign.

Answer: ________________ gallons per minute

5. The expression below is a derivative in disguise:

$$\lim_{h \to 0} \frac{(5+h)^3 - 125}{h}$$

Name the function f and the point a for which this is $f'(a)$, explain in one sentence how you recognised them, then evaluate the limit.

Answer: ________________

6. Let $f(x) = \dfrac{1}{x}$. Use the limit definition to find $f'(2)$. Combine the two fractions in the numerator over a common denominator before you cancel the h; write the value as a decimal.

Answer: ________________

7. A runner's distance from the start line is measured at three times, with no formula available.

t (seconds)	2	3	4
distance (metres)	29	35	41

Estimate the runner's instantaneous velocity at $t = 3$ by using the window that is centred on $t = 3$, and say in one sentence why that centred window is a better estimate than either one-sided window.

Answer: ________________ meters per second

8. Take any linear function $f(x) = mx+k$. Using only the limit definition, explain why the instantaneous rate of change is the same at every point, and why no table of average rates for a linear function ever needs a limit at all. Then state the units of $V'(t)$ from problem 3 and explain where those units come from in the difference quotient. Write in complete sentences.